MIL-P-18612A (OS)
3 January 1975

SUPERSEDING
MIL-P-18612 (NORD)
13 June 1955
Amendment 3
7 August 1972

MILITARY SPECIFICATION

PRIMER MARK 101 MOD 3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification governs the manufacture, assembly and preparation for delivery of Primer Mark 101 Mod 3, and the methods of inspection and test upon which acceptance will be based. Primer Mark 101 Mod 3 is percussion initiated.

2. APPLICABLE DOCUMENTS

2.1 The following standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 1390

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DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

LD 275908	Primer Mark 101 Mod 3
LD 166537	Test Set (Primer) Mark 135 Mod 0
LD 168793	Accessories, Detonator Container Mark 3 Mod 0
LD 168794	Accessories, Detonator Container Mark 2 Mod 0
553214	Firing Pin
553491	Test Ball
660787	Detonator Container Mark 3 Mod 0
660788	Detonator Container Mark 2 Mod 0
894013	Adapter Assembly

and all documents listed on the above LD's and drawings.

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

OD 5823	Sensitivity Test of Primers and Detonators using Test Set Mark 135 Mod 0 (Primer) and Test Set Mark 136 Mod 0 (Detonator)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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Code of Federal Regulations

49 CFR 1 to 199 Title 49 - Transportation
Parts 100 to 199

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 General. - Primers Mark 101 Mod 3 covered by this specification shall be manufactured in accordance with Naval Ordnance Systems Command List of Drawings LD No. 275908 and all documents listed thereon.

3.2 Primer mixture. - The mixture of ingredients specified on the applicable drawing shall be a thoroughly mixed, uniform powder.

3.3 Moisture content (M101). - The moisture content of the priming mixture at the time of loading shall be less than 0.30 percent as determined by tests specified in 4.5.1.

3.4 Loading. - The priming mixture shall be loaded by ram force as specified on the applicable drawings. Loading and assembly operations shall be carried out under conditions which will keep the moisture content of the explosive charges in the primers less than 0.30 percent by weight. The completed primer assembly shall be kept under dry storage conditions prior to packaging as specified in 5.1.

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3.5 Primer sensitivity (M102). - The sensitivity of the primer to firing pin impact shall be determined when tested as specified in 4.5.2. The sensitivity value shall be used for data card record and process control. Deviations in the sensitivity value shall not directly cause the rejection of primers, but may be reason for checking materials and processes for compliance with drawing and specification requirements.

3.6 Primer functioning (M103). - The primer shall function on firing pin impact when tested in Test Set (Primer) Mark 135 Mod 0 using a four ounce test ball and a drop height of 9.5 inches. Testing and acceptance will be in accordance with 4.5.3.

3.7 Simulated delay element functioning of primer (M104). - The primers shall function without blowback or rupture of the primer cup under the simulated delay element functioning test specified in 4.5.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted for inspection will be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. - As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

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4.1.2 Responsibility for inspection. - Unless otherwise specified in the contract or order, the contractor is responsible for the performance of all inspection requirements as specified herein. The contractor may use his own or any other facilities approved by the Government for the performance of inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Product inspection. - The mandatory inspection requirements for the product are specified by the classified characteristics in this specification and on the referenced drawing and also by the supplemental quality assurance provisions of the contract.

4.3 Preproduction sample. - Before entering into quantity production, a preproduction sample of six hundred (600) Primers Mark 101 Mod 3 shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These primers shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.3.1 Preproduction sample for a subsequent contract. The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

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4.4 Production Lot. - For procurement purposes each lot of loaded primers shall consist of 10,000 units plus those required for all test purposes. Unless otherwise stated in the contract, order, or requisition, all units for test purposes shall be at the expense of the Contractor. The production lot should not be confused with the lot for inspection purposes as defined in 4.1.1.

4.5 Tests.

4.5.1 Moisture content test. - Conduct at least one test for each lot of primers and such additional tests as the Inspector considers warranted by variations in the process and materials. Collect an approximate two gram sample of material being loaded in a tared, stoppered container at the site of and immediately prior to consolidating the charge in the primer. Weigh the container; open and expose the sample for drying at 50 $\pm$ 2 degrees C and at least 28 inches (Hg) vacuum until constant weight is reached. Cool the container in a desiccator; stopper and weigh. All weighings shall be accurate to $\pm$ 0.5 milligram. The presence of moisture in excess of 0.30 percent shall cause suspension of the charge loading until permanent corrective measures satisfactory to the Inspector have been completed. Primers which have been loaded with defective charges shall be rejected.

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4.5.2 Primer sensitivity test. - Select 100 primers from each lot for the Bruceton (Staircase) Sensitivity test employing 0.50 inch increments. Use Test Set (Primer) Mark 135 Mod 0, Naval Ordnance Systems Command List of Drawings LD No. 166537, with Test Ball, Naval Ordnance Systems Command Drawing 553491-2, Firing Pin, Naval Ordnance Systems Command Drawing 553214-2 and Adapter Assembly, Naval Ordnance Systems Command Drawing 894013. Test set operation and test procedures are described in OD 5823. Record, on the Ammunition Data Card specified in 5.4, the 50 percent firing height (h) and the standard deviation (s) calculated from the test data.

4.5.3 Primer functioning test. - Samples shall be selected and tested from each lot in accordance with the Multiple Sampling Plan, Inspection Level II, AQL 0.15 percent defective as specified in Standard MIL-STD-105. The initiation of the primer shall be accomplished by the use of Test Set Mark 135 Mod 0 equipped with the test ball specified in 4.5.2. Drop height shall be 9.5 inches. Primers which do not explode on the first impact of the firing pin shall be classed as defective. Disposition of the lot shall be in accordance with the sampling plan given above.

4.5.4 Simulated delay element functioning test of Primer. - The primers shall be mounted in the Blow Back Adapter Assembly NAVORD Drawing 1125125. This assembly consists of Adapter Body NAVORD Drawing 1125126, Adapter Base NAVORD Drawing 1125127 and Adapter Top NAVORD Drawing 894015, all of which are part of NAVORD LD 166537. The loaded adapter shall be mounted in the plate of Test Set (Primer) Mark 135 Mod 0 and the primers shall be subjected to functioning impacts as specified in 4.5.2. In this test, a fixed test ball height of 20 inches shall be used. A primer, which does not function or in which the cup ruptures or blows back, shall be considered defective. The test shall be conducted on the basis of a minimum sample of 50 items

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with acceptance where no defective is obtained. The incidence of one defective primer in 50 shall cause continued testing and disposition of the lot in accordance with the quality control methods specified in Standard MIL-STD-105 and the assigned quality level.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Primer Mark 101 Mod 3 shall be packaged in accordance with Naval Ordnance Systems Command List of Drawings LD No. 168793.

5.2 Packing. - Primer Mark 101 Mod 3 packaged as specified in 5.1 shall be packed in accordance with Naval Ordnance Systems Command List of Drawings LD No. 168794.

5.3 Marking. - In addition to any special markings required by the contract or order, all containers shall be marked in accordance with Naval Ordnance Systems Command Drawing Nos. 660787 and 660788, Standard MIL-STD-129, and Code of Federal Regulations 49 CFR 100 to 199.

5.4 Data cards. - Data cards shall be filled out for each lot of primers. Two of these cards shall accompany the shipping ticket; three shall be forwarded to the Procuring Agency and five shall be furnished the Inspector.

6. NOTES

6.1 Intended use. - The Primer Mark 101 Mod 3 is used as the initiating element in fuze explosive trains.

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6.2 Ordering data. - Procurement requirements should specify:

6.2.1 Procurement requirements. -

a. Title, number, and date of this specification.

b. The agency designated to test the preproduction sample (4.2).

c. The necessity for a preproduction sample (4.2.1).

d. Any special markings required (5.3).

e. That the safety precaution requirements of the Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4145.26, are applicable.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammunition Ashore, OP 5, should be made applicable.

6.2.2 Contract data requirements. - Items of deliverable data required by this specification are cited in the following paragraphs herein:

Paragraph	Data Requirement (Short Title)	Applicable DID
5.4	Ammunition Data Cards	DI-E-2001

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DIDS (Data Item Descriptions/DD Forms 1664) for the above data requirements are/will be (*) documented in the applicable ADL (Authorized Data List).

Such data will be delivered as identified on completed (numbered) DIDS when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

Preparing Activity:
Navy - OS
(Project No. 1390-N159)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-P-18612A(OS) Primer Mark 101 Mod 3

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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